



BUSINESS ANALYSIS

Lecture 1: Introduction to Business Analysis

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BUSINESS ANALYSIS COURSE SCOPE

- Business Analysis: Introductory Concepts
- Types of Business Analysis
- Requirements Engineering
- Stakeholder Management
- Requirements Elicitation
- Process Analysis & Process Modelling
- Evidence-Based Decision Making
- Key Performance Indicators and Metrics
- Business Data Analysis & Visualization
- Business Analysis in Software Development
- Ethical & Professional Practice

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TEACHING METHODOLOGY

- Lectures on business analysis
- Real-world case studies
- Student participation through presentations
- Learning methods
 - Skill-focused
 - Analytical focused
 - Groups and teams
 - Project-based

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EVALUATION METHOD

- Three Quizzes – 10%
- Three Short Assignments – 10 %
- Midterm Examination I – 15%
- Midterm Examination II – 15%
- Business-related Assignment & Report – 10%
- Final Examination – 40%

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BUSINESS ANALYSIS & BUSINESS ANALYST

- **What is business analysis?**

- “The practice of enabling change in an organizational context, by defining needs and recommending solutions that deliver value to stakeholders.”
- “A disciplined approach for introducing and managing change to organizations, whether for-profit businesses, governments, or non-profits.”

- **What does a business analyst do?**

Identifies and provides solutions that will maximize the value delivered by an organization to its stakeholders

- **Other titles used for business analyst**

- Business Systems Analyst / Systems Analyst
- Requirements Engineer
- Process Analyst
- Business Architect
- Business Intelligence Analyst
- Data Scientist

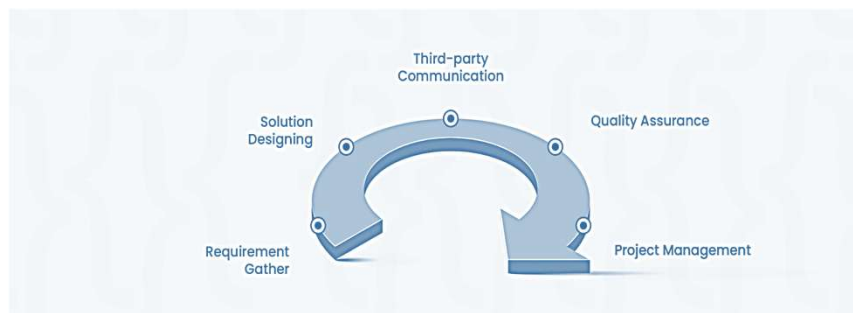
Source: IIBA; available at <https://www.iiba.org/professional-development/career-centre/what-is-business-analysis/>

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BUSINESS ANALYSTS ROLE IN SOFTWARE HOUSES

Role of a business analyst in the software development life cycle



Source: Programmers Force; available at <https://pf.com.pk/blogs/decode-the-role-a-comprehensive-guide-to-business-analyst-career-path/>

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BUSINESS ANALYSIS vis-à-vis BUSINESS ANALYTICS

What is Business Analytics

“Business analytics refers to the statistical methods and computing technologies for processing, mining and visualizing data to uncover patterns, relationships and insights that enable better business decision-making” (IBM)

“Business analytics is the process of using quantitative methods to derive meaning from data to make informed business decisions” (Harvard Business School)

Sources:
IBM; available at <https://www.ibm.com/think/topics/data-driven-decision-making>
Harvard Business School; available at <https://online.hbs.edu/blog/post/importance-of-business-analytics>
Northeastern University; <https://graduate.northeastern.edu/knowledge-hub/what-does-a-business-analyst-do/>

Similarities, Overlaps and Differences

- “Business analysis” and “business analytics” are often used interchangeably
- Both involve the review of data-based information to benefit a corporation
- The unique qualities of the two disciplines:
 - Business analysis focuses on process improvement and solution implementation
 - Business analytics is based on data and involves using that data to formulate conclusions on business performance.
- The lines between these disciplines continue to be blurred as industries become reliant on data for high-level decision-making

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BUSINESS ANALYSIS vis-à-vis BUSINESS ANALYTICS

	Business Analysis	Business Analytics
Focus	Processes, techniques, functions	Data and statistical analysis
Use	Address complex business problems	Predict future states & drive business decisions
Performed by	Business Analyst, Systems Analyst etc.	Data Analyst, Data Scientist etc.
Skills required	Analysis, elicitation, modelling, documentation	Statistics, mathematics, software programming etc.
Areas of impact	Organization, Process, Business, Technology	Business decisions, data-driven decisions, forecasting etc.

Source: The Business Analyst; available at <https://thebusinessanalystjobdescription.com/difference-between-business-analysis-and-business-analytics/>

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BUSINESS PROBLEMS vis-à-vis TECHNICAL PROBLEMS

Business Problems

- Business problems focus on the "why" e.g.
 - Why have profits decreased
 - Why are customers complaining about a service
- Business problems are about the goals of a business

Technical Problems

- Technical problem focuses on the "what," e.g.
 - What process malfunctions is creating decline in profits
 - What software defect is effecting service delivery
- Technical problems are about the underlying root causes that need to be addressed to reach goals.

Source: a SALESGROWTH Company; available at <https://salesgrowth.com/solve-technical-problems-or-business-problems/>

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STARTING WITH REAL WORLD EXAMPLES

(SOURCE: Harvard Business School (<https://online.hbs.edu/blog/post/business-analytics-examples>))

MICROSOFT: How to increase collaboration among employees and increase their productivity?

- Assumption (hypothesis): People will collaborate more when they are closely located
- Intervention (decision taken by management): Move 1200 employees from 5 buildings to 4
- Business analysis: Data/information used for evaluating the outcome of intervention
 - Travel time for meetings
 - No. of meetings
- Result of interventions
 - Travel time for meetings reduced by 46%: 100 work hours saved per week for 1200 employees resulting in annual savings of \$520,000
 - No. of meetings per week per person increased from 14 to 18
- Conclusion reached by business analysts: "Thoughtful planning of employee workspaces could lead to significant time and cost savings"

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STARTING WITH REAL WORLD EXAMPLES

(SOURCE: Harvard Business School (<https://online.hbs.edu/blog/post/business-analytics-examples>))

UBER: How to enhance customer support?

- Assumption (hypothesis): Improving Uber's online system for customer support, COTA v1 (Customer Obsession Ticket System) will increase the efficiency with which UBER agents identify and resolve issues
- Intervention (decision taken by management): Development of COTA v2
- Method used by business analysts to evaluate the outcome of intervention:
 - Selection of a group of customers (comprising sub-groups A & B)
 - Exposing both A & B first to COTA v1 for one week
 - Exposing sub-group B (treatment group) to COTA v2 for a month
 - Measuring the efficiency of COTA v1 vis-à-vis COTA v2 on the treatment group
- Result of interventions analyzed by the analysts at Uber
 - Average time to handle an issue reduced by seven percent for COTA v2
- Conclusion reached by business analysts: "implementing COTA v2 would not only improve customer service, but save millions of dollars by streamlining its ticket resolution process"

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STARTING WITH REAL WORLD EXAMPLES

(SOURCE: Harvard Business School (<https://online.hbs.edu/blog/post/business-analytics-examples>))

PepsiCo: How to maximize profits by ensuring that right quantities and types of products are available to consumers in certain locations?

- Intervention (decision taken by management): Development of cloud-based data and analytics platform Pep Worx
- Example of how Pep Worx used "big data" to maximize profits by data-driven decision-making
 - Use of data of 110 million households (big data) to identify potential customers for the Company's product *Quaker Overnight Oats*
 - Identification of 24 million households out of this data interested in the product
 - Intervention: Identification and targeting of retailers where these 24 million households shopped
 - Result of intervention: 80% growth of the product sales driven by these 24 million households

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INTRODUCING SOME KEY CONCEPTS USED IN THE FORE-GOING EXAMPLES

Term	Meaning
Business	An organized economic activity that provides goods or services mostly in exchange for payments and is driven by profit or some desired social outcome
Hypothesis	A statement indicating a cause-effect relationship based on observation and testable through research or experimentation
Intervention	An action/step/decision taken by a company to achieve an objective or objectives
Data	Facts or figures which can be processed and analyzed to extract meaningful information
Data-driven Decisions	Data-driven decision-making (DDDM) emphasizes using data (such as customer feedback, market trend) and analysis instead of intuition to inform business decisions ⁽²⁾

Source: IBM; available at <https://www.ibm.com/think/topics/data-driven-decision-making>

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TEN MINUTES ACTIVITY

1. Identify a business of interest
2. Draft a hypothesis of an area of interest in that business; examples:
 - If A is done, B will happen;
 - A is a pre-condition for B to happen
3. Suggest the intervention "A"
4. What kind of data/information is required to assess the effectiveness of intervention "A"
5. Can you suggest what methods can be used to analyze the data? If you can, you get a star



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WHY BUSINESS ANALYSIS

- Improved efficiency and productivity
- Faster, more effective decision-making
- Better financial performance
- Identification and creation of new revenue streams
- Improved customer acquisition and retention

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INTRODUCTION TO CORE BUSINESS FUNCTIONS

“Core business function:

Activities of an enterprise yielding income: the production of final goods or services intended for the market or for third parties. Usually, the core business functions make up the principal activity of the enterprise, but they may also include other (secondary) activities if the enterprise considers these as part of its core functions.

Support business function:

Supporting activities carried out by the enterprise in order to permit or to facilitate the core business functions, its production activity. The outputs (results) of support business functions are not themselves intended directly for the market or for third parties.”

Source: Eurostat, *Statistics explained*; available at https://ec.europa.eu/eurostat/statistics-explained/index.php/Glossary:Business_functions

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INTRODUCTION TO CORE BUSINESS PRACTICES & FUNCTIONS

Functional areas of a business

“Businesses often split their work into different departments or functional areas. Each of these functional areas will have specific tasks to complete. The main functional areas are:

- marketing
- human resources
- operations
- finance”

Source: BBC, *Business in action*; available at <https://www.bbc.co.uk/bitesize/guides/zpww3k7/revision/1>

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INTRODUCTION TO CORE BUSINESS PRACTICES & FUNCTIONS

Marketing

“The marketing department is concerned with finding out the needs and wants of potential customers and promoting the company’s products or services.

The main functional activities carried out by the marketing department are:

- advertising
- market research
- promotion
- selling products”

Examples of marketing data

- Social Media: Engagement (data on likes, shares), reach, follower growth
- Customer Data: Demographics, purchase history, preferences

Examples of marketing data analysis

- Targeting potential customers for a product using data of purchase history
- Using customer surveys to offer products tailored to specific needs
- Using existing data on sales and market trend to predict future sales and plan output accordingly

Source: BBC, *Business in action*; available at <https://www.bbc.co.uk/bitesize/guides/zpww3k7/revision/1>

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INTRODUCTION TO CORE BUSINESS PRACTICES & FUNCTIONS

Human Resource

"The human resources department manages the people who work for the organisation. The main functional activities carried out by the human resources department are:

- recruitment and selection
- staff training
- industrial relations
- health and safety"

Examples of HR data

- Recruitment data such as time taken to select employees
- Employee productivity data
- Data of employees' qualification

Examples of HR data analysis

- Employees' performance over time
- Salary structure and impact on employees' performance
- Employee profile/ features and job placement

Source: BBC, *Business in action*; available at <https://www.bbc.co.uk/bitesize/guides/zpww3k7/revision/1>

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INTRODUCTION TO CORE BUSINESS PRACTICES & FUNCTIONS

Operations

"The operations department is responsible for making the product or providing the service. It is their responsibility to ensure that the product or service is of the required quality to satisfy the customer.

The main functional activities carried out by the operations department are:

- production of goods or services
- buying
- raw materials
- control of quality"

Examples of operations data

- Data on use of raw materials in the manufacturing process
- Time spent on different business tasks
- Time to answer online customer queries

Examples of operations data analysis

- Identifying cheap sources for raw materials
- Predicting how a new process could reduce time spent on a business task
- Diagnosing the effectiveness of quality control systems

Source: BBC, *Business in action*; available at <https://www.bbc.co.uk/bitesize/guides/zpww3k7/revision/1>

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INTRODUCTION TO CORE BUSINESS PRACTICES & FUNCTIONS

Finance

"The finance department is responsible for managing the money coming into and going out of the business. They will deal with all the financial aspects of the business such as paying suppliers, paying staff wages and receiving payments from customers.

The main functional activities carried out by the finance department are:

- raising finance
- preparing budgets
- preparing final accounts"

Examples of financial data

- Profit and loss statement: Profit, loss, expenses
- Balance sheet: Assets, liabilities
- Market & Economic data: Stock price, exchange rate, inflation, price of inputs used in production (e.g., oil, raw material)

Examples of financial data analysis

- Identifying or predicting areas where business expenses have increased or may increase
- Identifying products which have yielded higher profit margins
- Identifying how economic factors such as inflation or exchange rate may impact business profitability

Source: BBC, *Business in action*; available at <https://www.bbc.co.uk/bitesize/guides/zpww3k7/revision/1>

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STUDENT PRIORITY AREAS

Question 1: Level of business engagement:

- Entrepreneur
- Professional

Question 2: Kind of business firm

- Small or medium enterprise
- Multinational
- Non-profit
- Government

Question 3: Sector of preference; e.g.,

- Food
- Construction material

Question 4: Field of interest

- Marketing
- Finance
- Operations
- Administration

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ESSENTIAL READINGS

Harvard Business School

Page Title: Business Insights: Examples of business analytics in action

Available at: <https://online.hbs.edu/blog/post/business-analytics-examples>

BBC

Page Title: Business in action

Links

1. [Functional areas](#)
2. [Marketing](#)
3. [Human resources](#)
4. [Operations](#)
5. [Finance](#)
6. [Functional areas working together](#)

eurostat

Page Title: Glossary: Business functions

Available at: https://ec.europa.eu/eurostat/statistics-explained/index.php?title=Glossary:Business_functions

United Nations Department of Economic and Social Affairs

Title: The concept of business functions and their use in statistical surveys (pages 7 to 10) in *Manual on the Classification of Business Functions*; December 2022

Available at: https://unstats.un.org/unsd/classifications/Econ/Download/Manual_on_CBF_pre-edited_9Jan2023.pdf

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ESSENTIAL READINGS

IIBA

Page Title: What is business analysis

Available at: <https://www.iiba.org/professional-development/career-centre/what-is-business-analysis/>

Programmers Force

Page Title: Decode the role

Available at: <https://pf.com.pk/blogs/decode-the-role-a-comprehensive-guide-to-business-analyst-career-path/>

Northeastern University

Page Title: What does a business analyst do

Available at: <https://graduate.northeastern.edu/knowledge-hub/what-does-a-business-analyst-do/>

The Business Analyst

Page Title: Business Analysis vs Business Analytics

Available at: <https://graduate.northeastern.edu/knowledge-hub/what-does-a-business-analyst-do/>

A SALESGROWTH COMPANY

Page Title: Technical Problems vs Business Problems

Available at: <https://salesgrowth.com/solve-technical-problems-or-business-problems/>

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SUGGESTED READINGS

Harvard Business School

Page Title: Business Insights: How to analyze a dataset: 6 steps

Available at: <https://online.hbs.edu/blog/post/how-to-analyze-datasets>

Great Learning (Blog)

Links:

[Definition of Data](#)

[Classification of Data](#)

[Importance of Data](#)

[How Do We Analyze Different Data?](#)

[Challenges in Data Management](#)

[Future Trends in Data Management](#)

[Conclusion](#)

[Frequently Asked Questions\(FAQ's\)](#)

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LECTURE 2

- Business analysis and business strategies
 - Total Quality Management (TQM)
 - Six Sigma
 - ISO 9000
- SWOT analysis for strategic planning

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